Thin Film Interference

- Bright colors on certain surfaces are caused by thin film interference - e.g.:

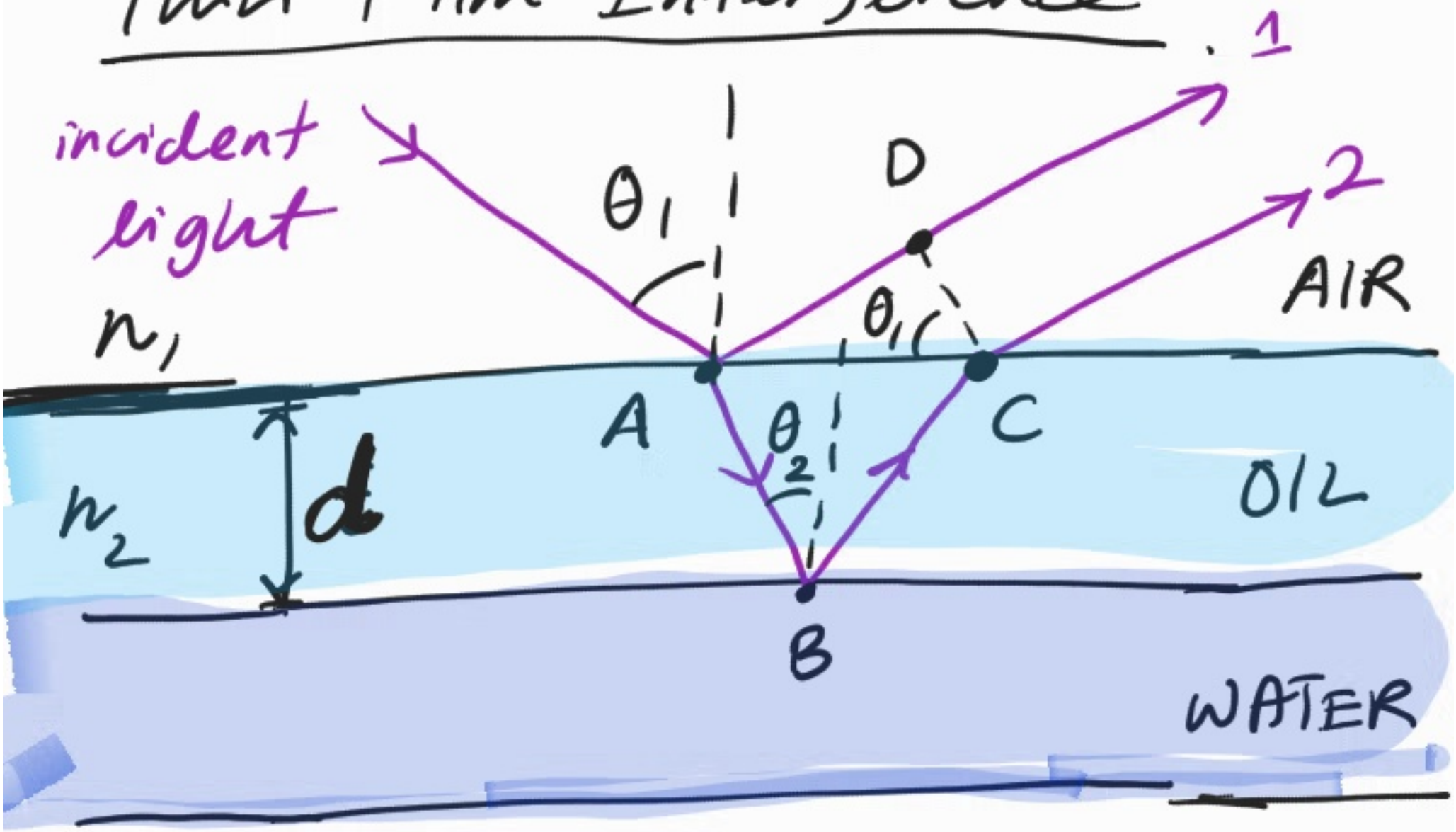
a) sunlit surface of a soap bubble
b) colorful patterns when light reflects off a compact disk.

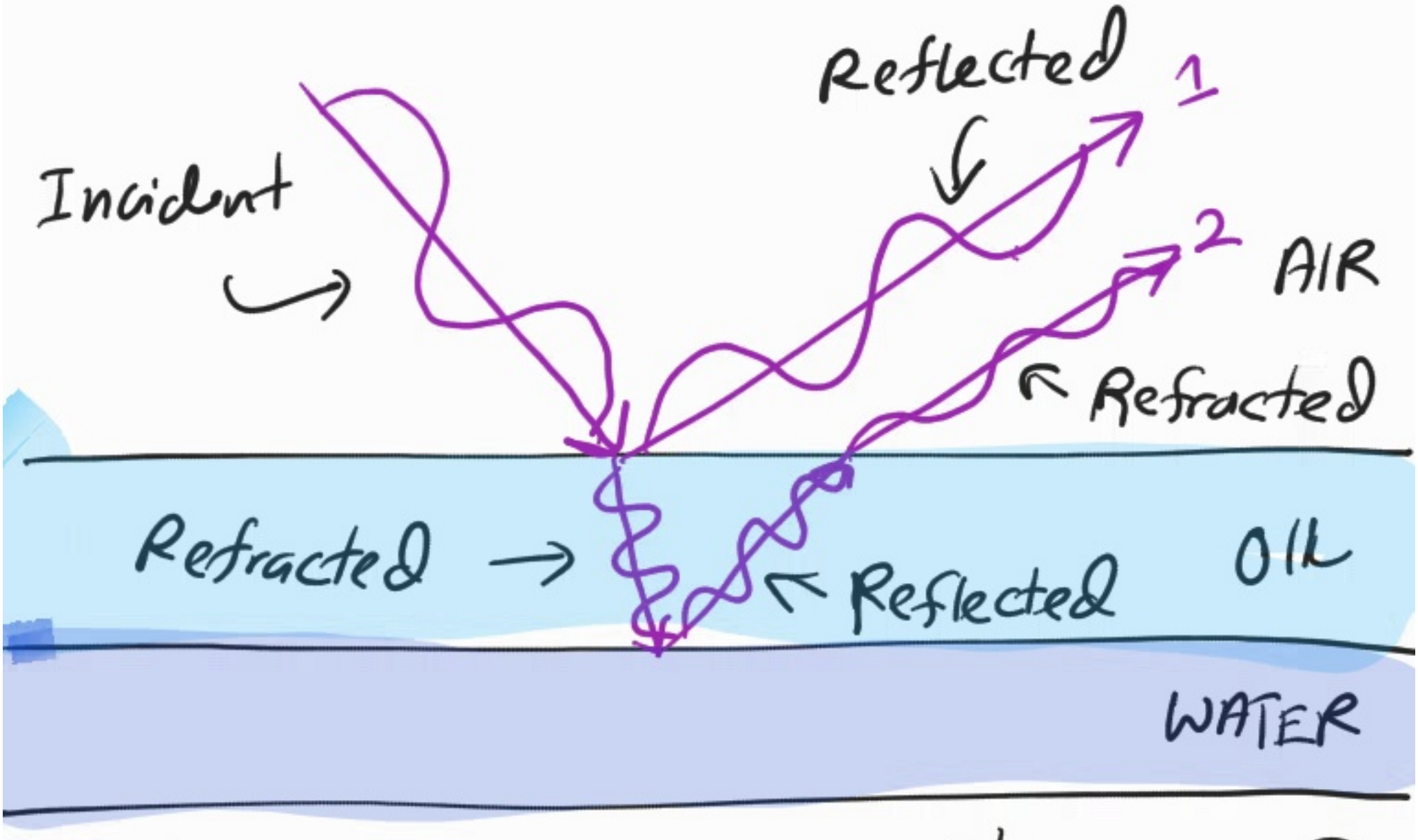
- This phenomenon occurs when light waves are reflected by the lower & upper boundaries of a thin film interferes with one another resulting in:

⇒ Contructive Interference
- enhancing reflected light

⇒ Destructive Interference
- reducing reflected light

Thin Film Interference

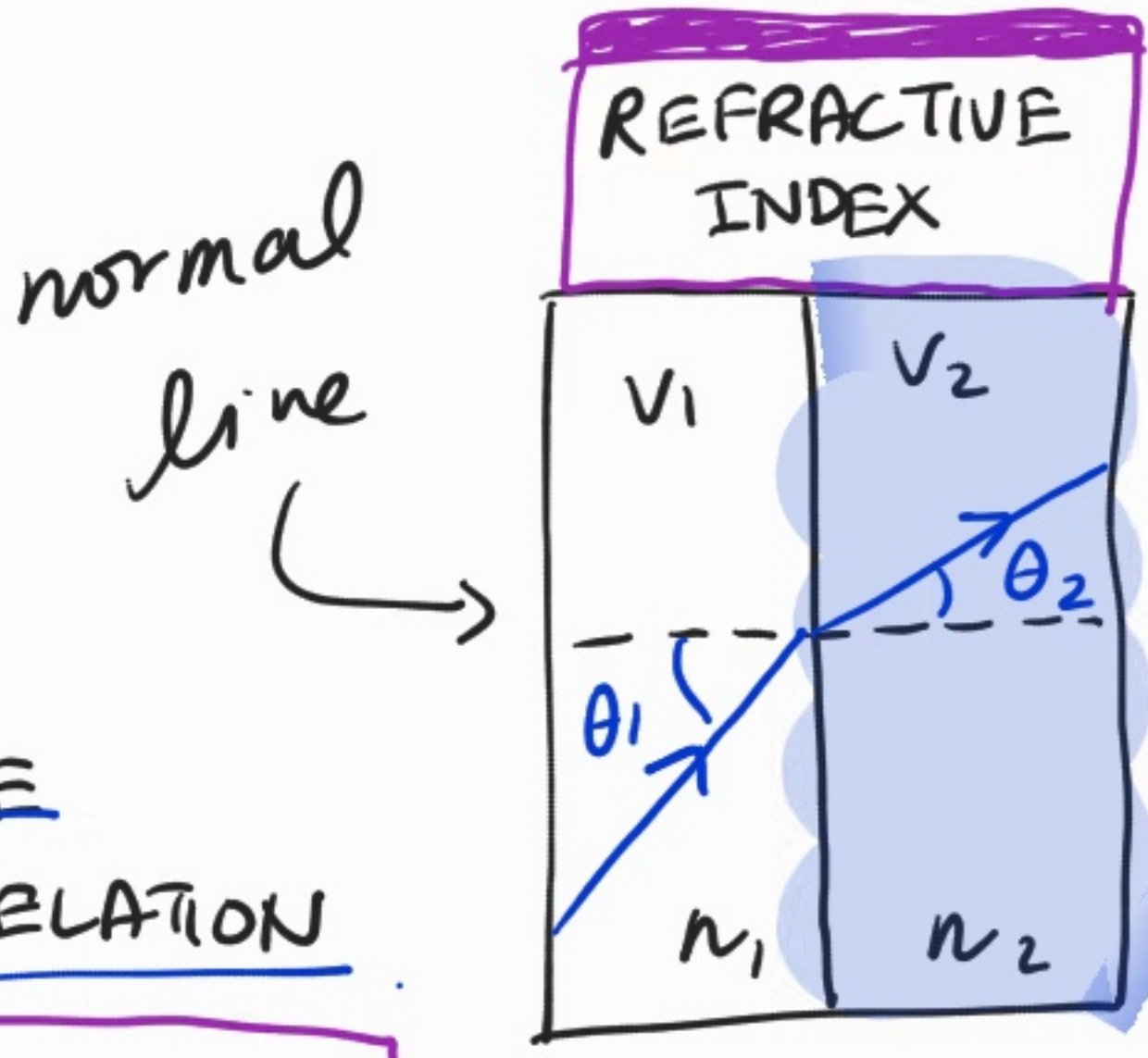




- Homogeneous assumption.

Thin Film Interference

Snell's Law:



REFRACTIVE INDEX RELATION

$$\frac{n_1}{n_2} = \frac{\sin\theta_2}{\sin\theta_1}$$

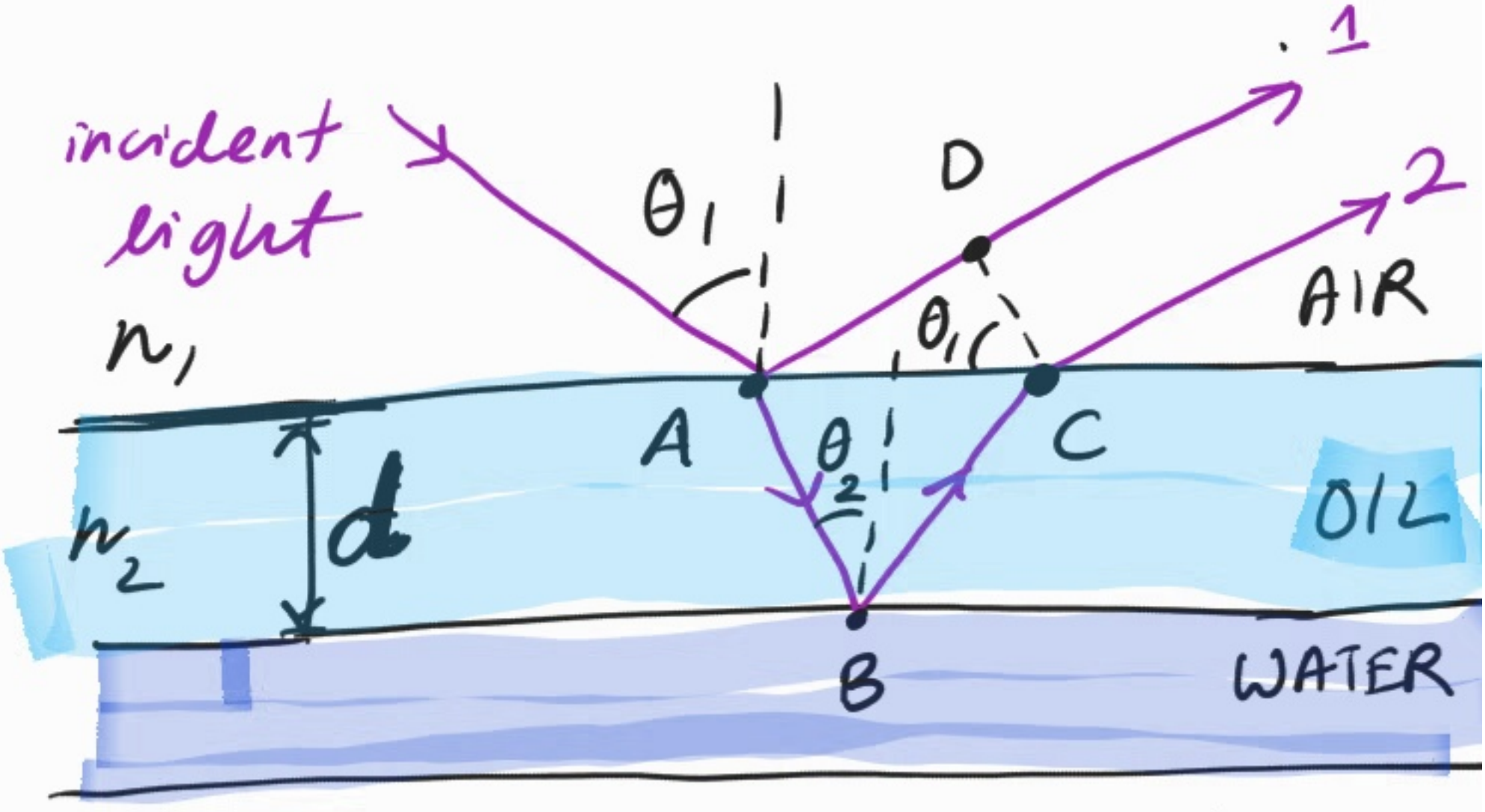
VELOCITY RELATION

$$\frac{V_1}{V_2} = \frac{\sin\theta_1}{\sin\theta_2}$$

Thin Film Interference

- Interference effects are most significant when light interacts with something having a size similar (or smaller) than its λ.

$[d < \lambda]$



- Ray 1 and Ray 2 may be in or out of phase.

Thin Film Interference

Consider a situation when the film has very small thickness, d; where the difference between the path lengths are negligible. Does a phase change occur?

Thin Film Interference

- Light waves undergo a phase change by π rad upon reflection at the interface if $n_2 > n_1$.

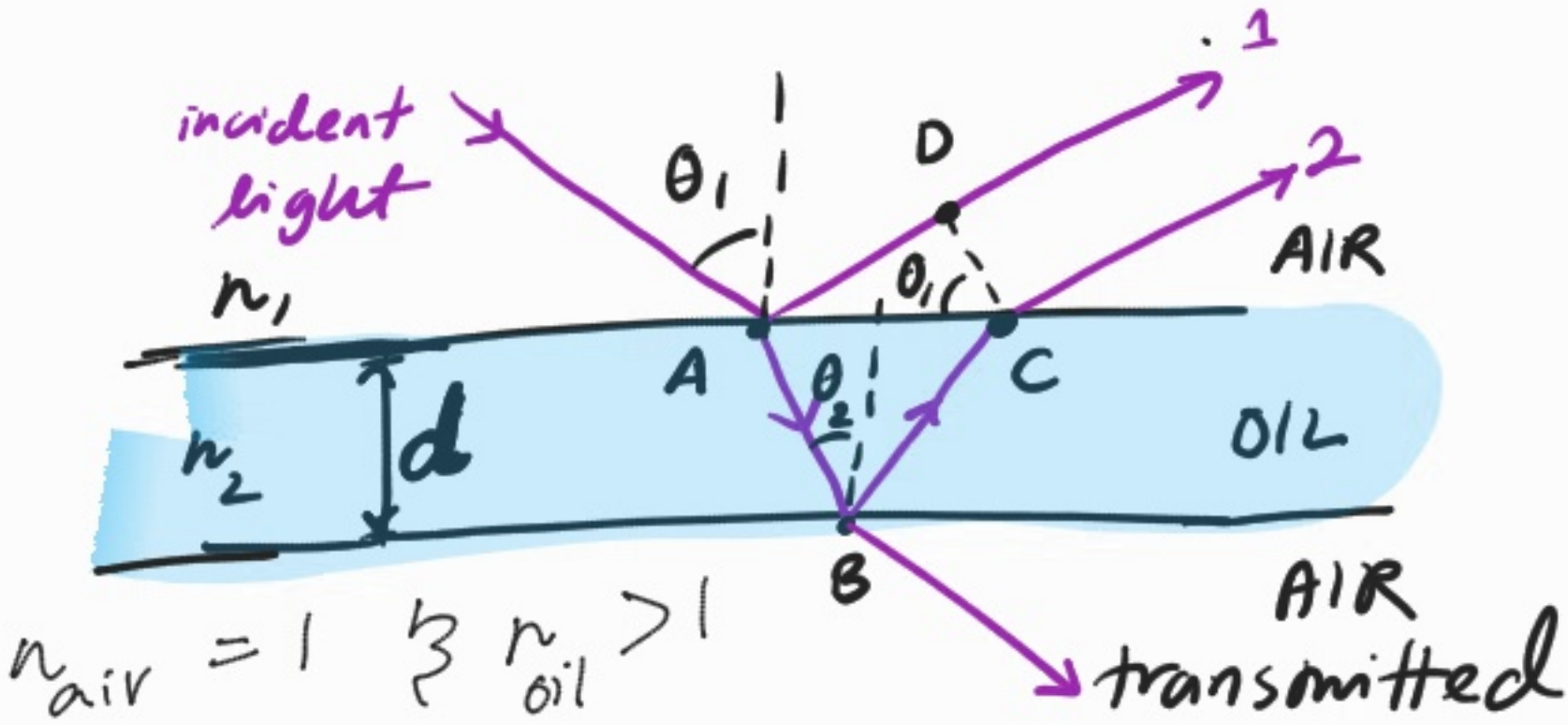
If the film is very thin, & the path length difference between the two rays are negligible :

→ then the 2 rays are exactly out of phase

→ destructive interference occurs at all λ of the incident ray

[DARK APPEARANCE]

Thin Film Interference



$n_{air} = 1 \;\; \& \;\; n_{oil} > 1$

⇛ reflection at the upper boundary (air-oil) would introduce a π rad (180°) phase shift in the reflected wave. ($n_{air} < n_{oil}$)

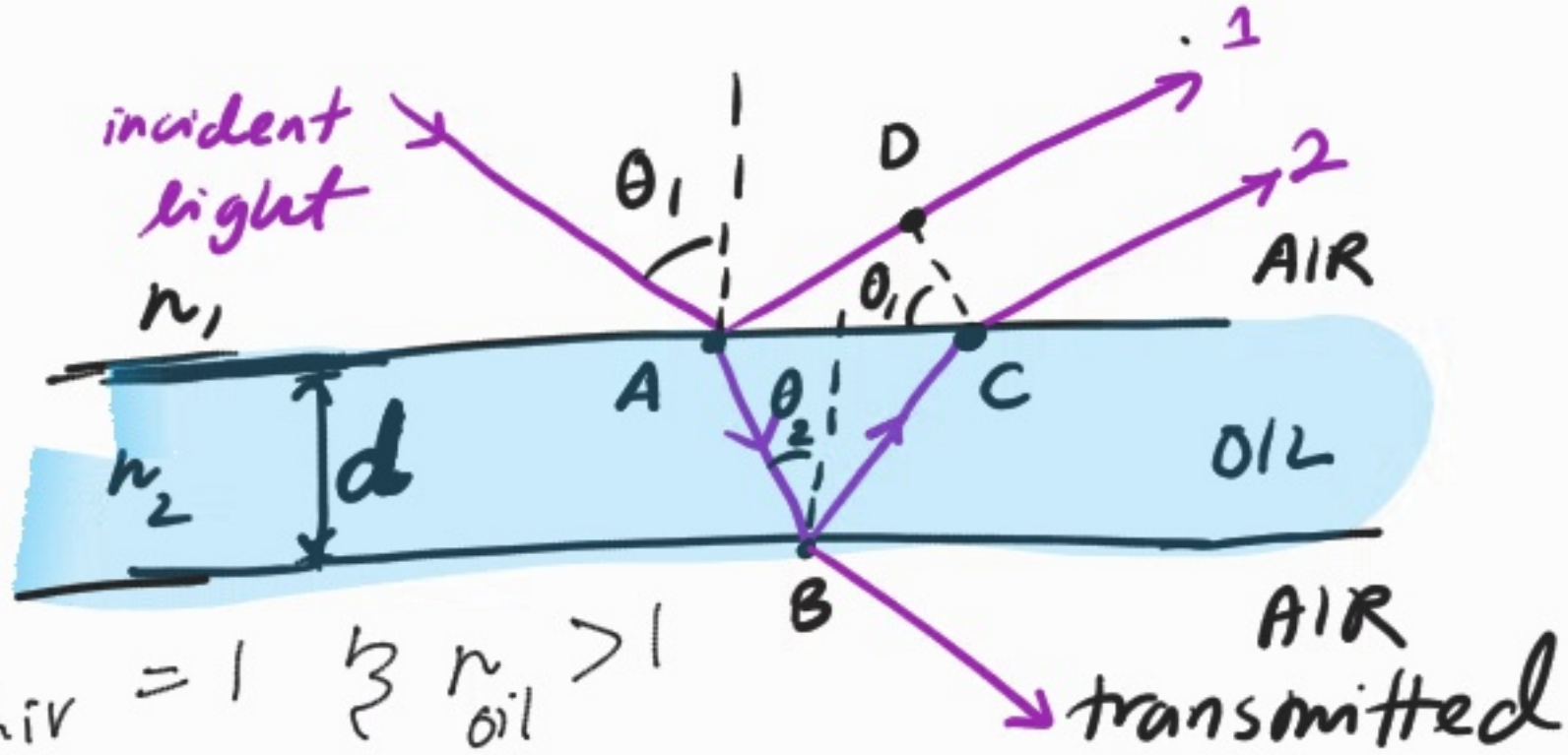
⇛ reflection at the lower boundary (air-oil) would **NOT** change the phase of the reflected wave → $n_{oil} > n_{air}$

Thin Film Interference

Optical Path length

product of geometric length of the optical path (light)(s) & the refractive index of a homogeneous medium (n);

$$OPL = ns$$

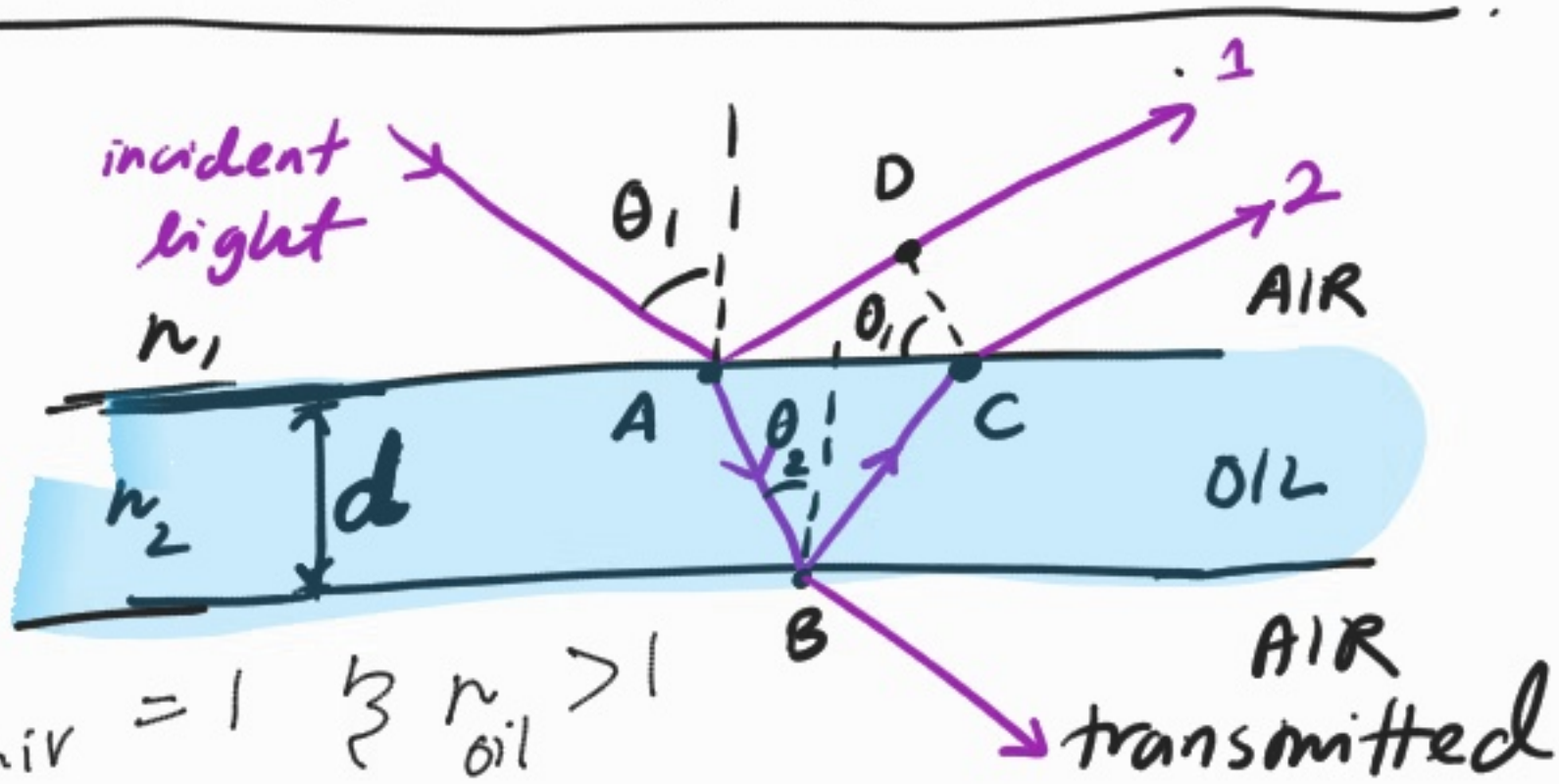


$n_{air} = 1$ & $n_{oil} > 1$

Optical path difference

$$OPD = \underbrace{n_2(\overline{BC} + \overline{AB})}_{\text{OIL}} - \underbrace{n_1(\overline{AD})}_{\text{AIR}}$$

Thin Film Interference



$n_{air} = 1 \; \& \; n_{oil} > 1$

$\overline{AB} = \overline{BC} = \dfrac{d}{\cos\theta_2} \; \& \; \overline{AD} = 2d \tan\theta_2 \times \sin\theta_1$

Snell's Law $\Rightarrow n_1/n_2 = \sin\theta_2/\sin\theta_1$

$$OPD = n_2\left[\frac{2d}{\cos\theta_2}\right] - 2d\tan\frac{\theta}{2}(n_2\sin\theta_2)$$

$$= 2n_2 d\left[\frac{1-\sin^2\theta_2}{\cos\theta_2}\right]$$

$$= 2n_2 d\cos\theta_2$$

Destructive Interference $\Rightarrow 2n_2 d\cos\frac{\theta}{2} = m\lambda$

Thin Film Interference

CONSTRUCTIVE INTERFERENCE

$$2 n_{oil} d \cos(\theta_2) = \left(m - \frac{1}{2}\right)\lambda$$

DESTRUCTIVE INTERFERENCE

$$2 n_{oil} d \cos(\theta_2) = m\lambda$$

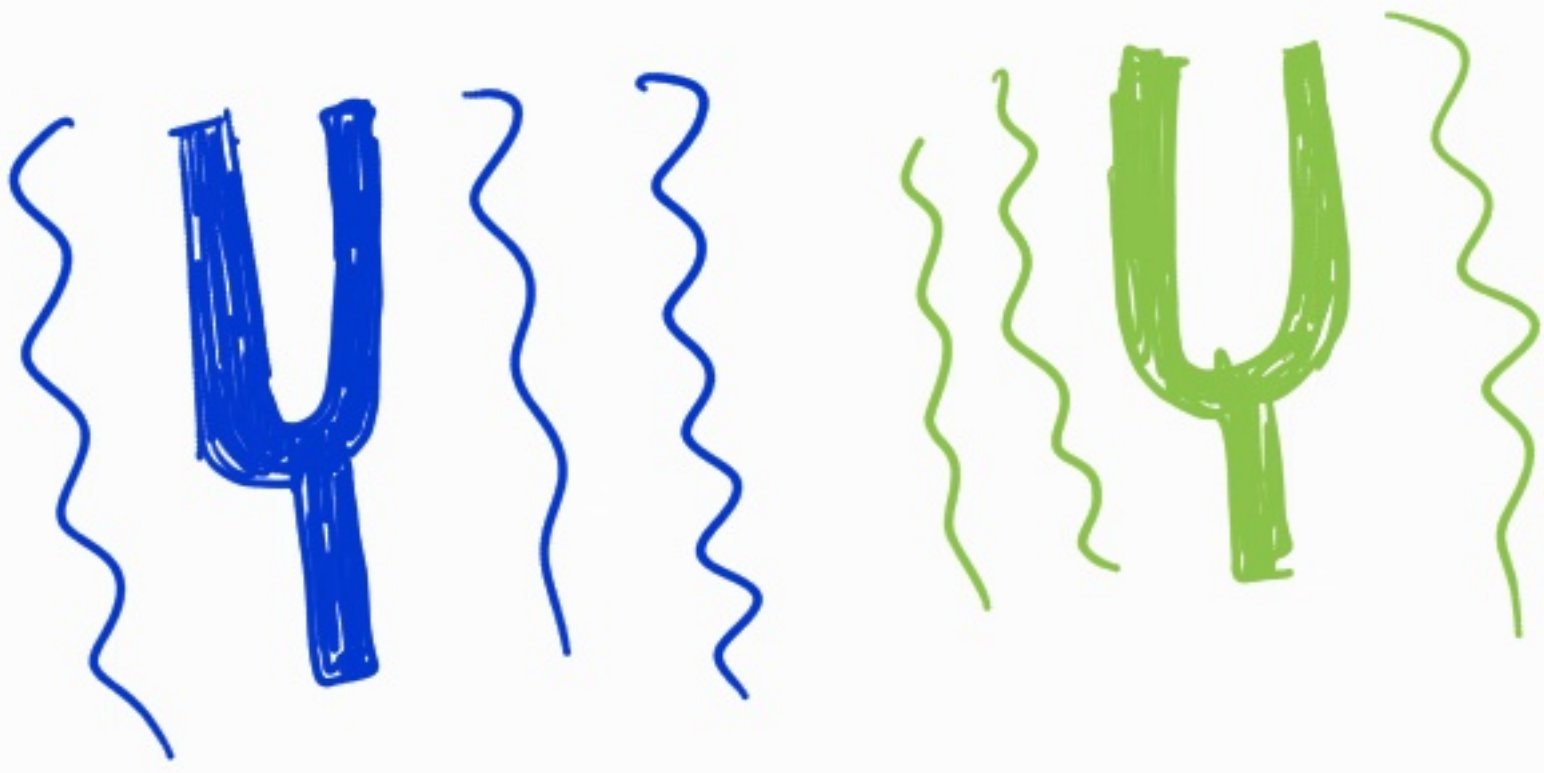
Note:

Interference pattern is dependent on :

a) wavelength of light, λ
b) film thickness, d
c) refractive indices on both sides of the film
d) refractive index of film.

Practice Problem [REVISION]

Consider 2 tuning forks: One has a frequency of 200 Hz and the other has a frequency of 400 Hz. If both forks are struck simultaneously, what is the beat frequency generated?



Solution.

- The beat frequency definition:
  ↳ absolute difference of the frequencies of both sound waves

$$\therefore f_{\text{beat}} = |f_1 - f_2|$$

$$= |200 - 400| \text{ Hz}$$

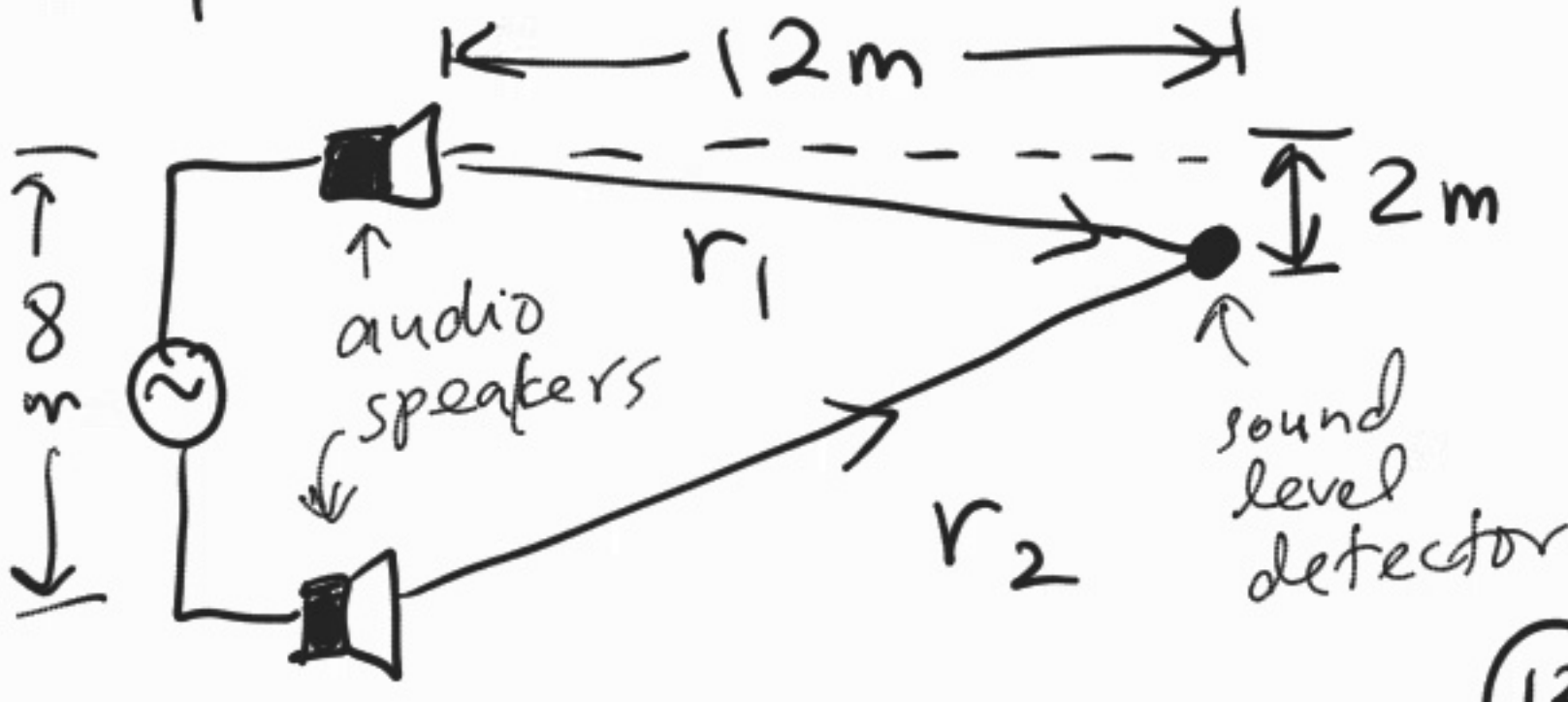
$$= 200 \text{ Hz}$$

$$\#\#$$

Practice Problem:

Sound Wave Interference.

Consider 2 audio speakers which generate a same signal (sound) at an unknown frequency. Both speakers are seperated by a distance of 8 m. If a sound level detector is placed as shown below, the 1st minimum intensity is found. Determine the frequency of the speakers' signal generator assumming the speed of sound is 343 m/s.



Strategy:

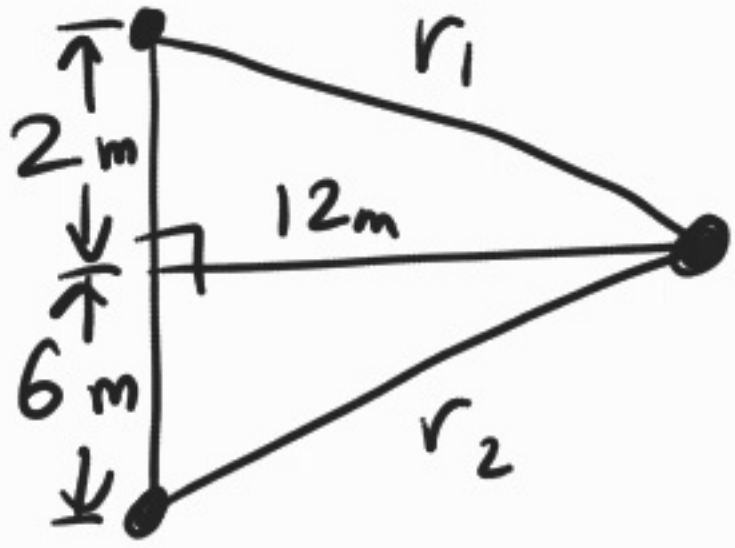
1, Calculate minimum path length between audio speakers and sound level detector.

2, Use the path length difference to determine the wave length.
(Destructive Interference)

3, Calculate frequency

$$f = \frac{v}{\lambda}$$

Solution.



⇒ Path length to the min point:

$$r_1 = \sqrt{12^2 + 2^2} = 12.17\,m$$

$$r_2 = \sqrt{6^2 + 12^2} = 13.42\,m$$

⇒ Path length difference to find λ

$$\Delta r = |r_2 - r_1| = 1.25\,m$$

⇒ 1st minimum intensity
= 1st destructive interference

$$\Delta r = |r_2 - r_1| = n\frac{\lambda}{2} \rightarrow \text{Set } n = 1$$

$$2\Delta r = \lambda$$

Solution [Con't]

$$\lambda = 2\Delta r = 2(1.25\,m)$$
$$= 2.5\,m$$

Frequency:

$$f = \frac{v}{\lambda} = \frac{343\ m/s}{2.5\ m} = 137.2\,Hz$$
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